

# Report on gearbox failure analysis

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## 1 Introduction

The data has been taken from 'Wind Turbine Gearbox Condition Monitoring Vibration Analysis Benchmarking Datasets' [1].

From 'Vibration Condition Monitoring Benchmarking Datasets.pdf' provided, following data has been used. Pages and tables have been referenced for convenience wherever needed.

The vibration data was collected at a 40 kHz sampling rate (p. 7). Data has been collected for a total of 10 minutes for both healthy and damaged state and has been divided into 10 equal parts - AN3, AN4, A5, A7, A8, A9, A10 (Figure 5, p. 6 for sensor placement).

Given,

Nominal High-Speed Shaft (HSS) speed of 1800 rpm (30 Hz) and a Main Shaft (LS) speed of 22.09 rpm (0.368 Hz) (Table 4, p. 8):

Characteristic frequencies (Table 1, p. 4)

- **High-Speed Stage (HS-ST):** Driven by a 22-tooth pinion and an 88-tooth gear (Gear Ratio: 4.0).
  - *Characteristic Driving Frequency ( $f_{HSS}$ ):*  $1800 \text{ RPM}/60 = 30 \text{ Hz}$ .
  - *Calculated Gear Mesh Frequency ( $GMF_{HS}$ ):*  $30 \text{ Hz} \times 22 \text{ teeth} = 660 \text{ Hz}$ .
- **Intermediate Speed Stage (IMS-ST):** Driven by a 23-tooth pinion and an 82-tooth gear (Gear Ratio: 3.57).
  - *Characteristic Shaft Frequency ( $f_{IMS}$ ):*  $30 \text{ Hz}/4.0 = 7.5 \text{ Hz}$ .
  - *Calculated Gear Mesh Frequency ( $GMF_{IMS}$ ):*  $7.5 \text{ Hz} \times 82 \text{ teeth} = 615 \text{ Hz}$ .

## 2 Initial Screening: RMS and CF

Each 1-minute discrete data vector was dynamically detrended by subtracting its local mean value to strip away electronic DC bias from the sensors. Following data cleanup, the Root Mean Square (RMS) acceleration energy which tracks overall operational power and the Crest Factor (CF) which measures signal impulsivity and sharp peak shocks—were extracted across eight casing sensors (AN3 to AN10).

From Figure 1, it is observed that the healthy baseline profiles remained entirely flat and low for all the cases. Meanwhile, the 'damaged' data exhibited a severe, permanent

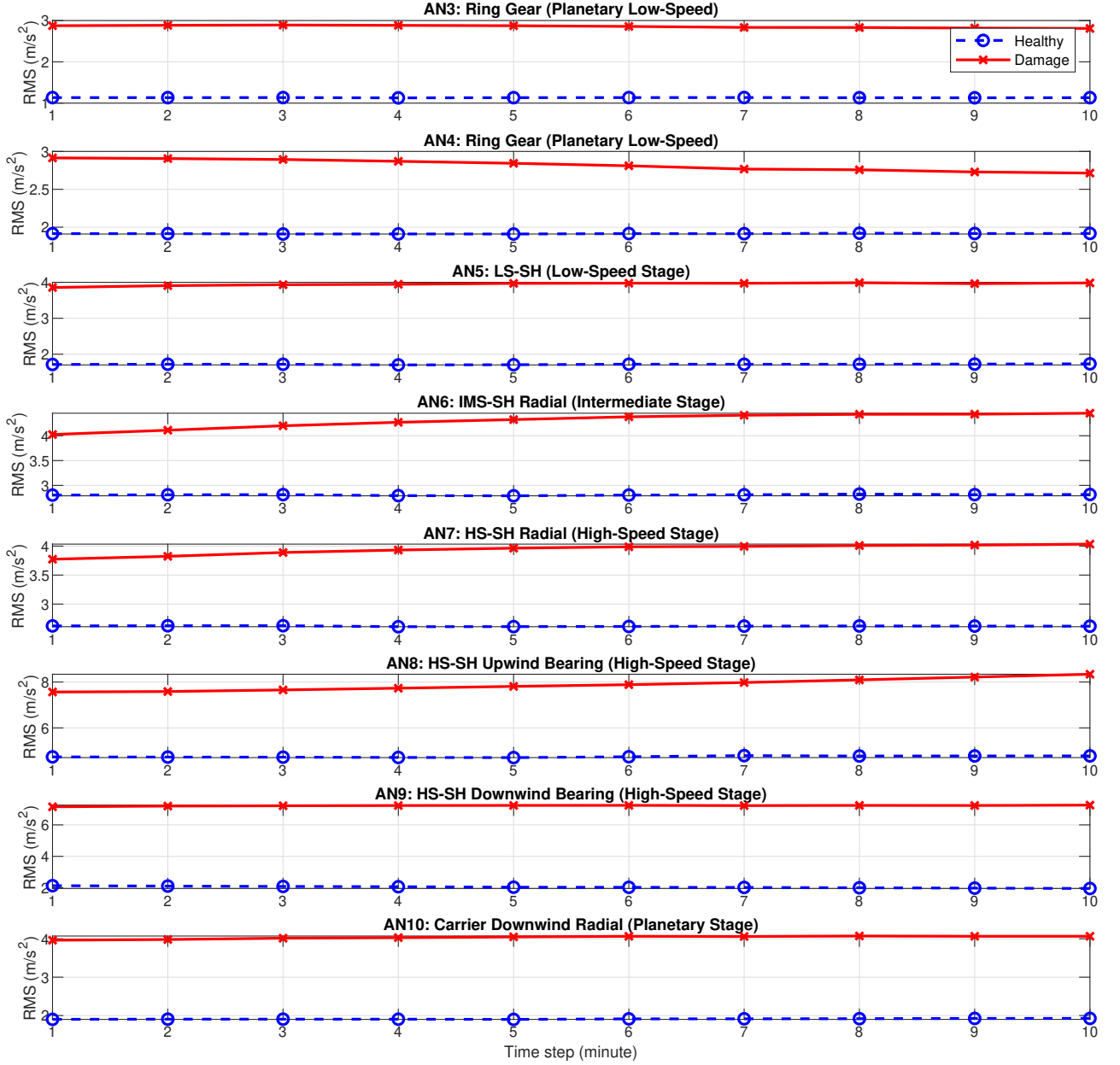


Figure 1: RMS for all acceleration data (Healthy vs. Damaged)

vertical energy offset from the 'healthy' benchmark across all sensors. This upward trend was most severe on sensors AN6 (Intermediate Parallel Stage Casing) and AN7 (High-Speed Parallel Stage Casing), both of which displayed a progressive linear upward gradient over time.

The permanent vertical energy gap in the RMS plots shows that a macro-structural damage has actively developed within the parallel speed meshing regions (AN6 and AN7).

For AN3 (Ring Gear) in Figure 2, the healthy baseline showed a highly impulsive trend, peaking at a CF of more than 8 at times. The damaged state trend line, however, was heavily suppressed to 4.2 to 4.9, remaining almost completely flat across the entire timeline.

The severe suppression of CF on AN3 provides critical insight into the wear mechanism. CF measures the ratio of sharp impact peaks relative to background continuous energy, a dropping or suppressed CF indicates that isolated, impulsive impact shocks

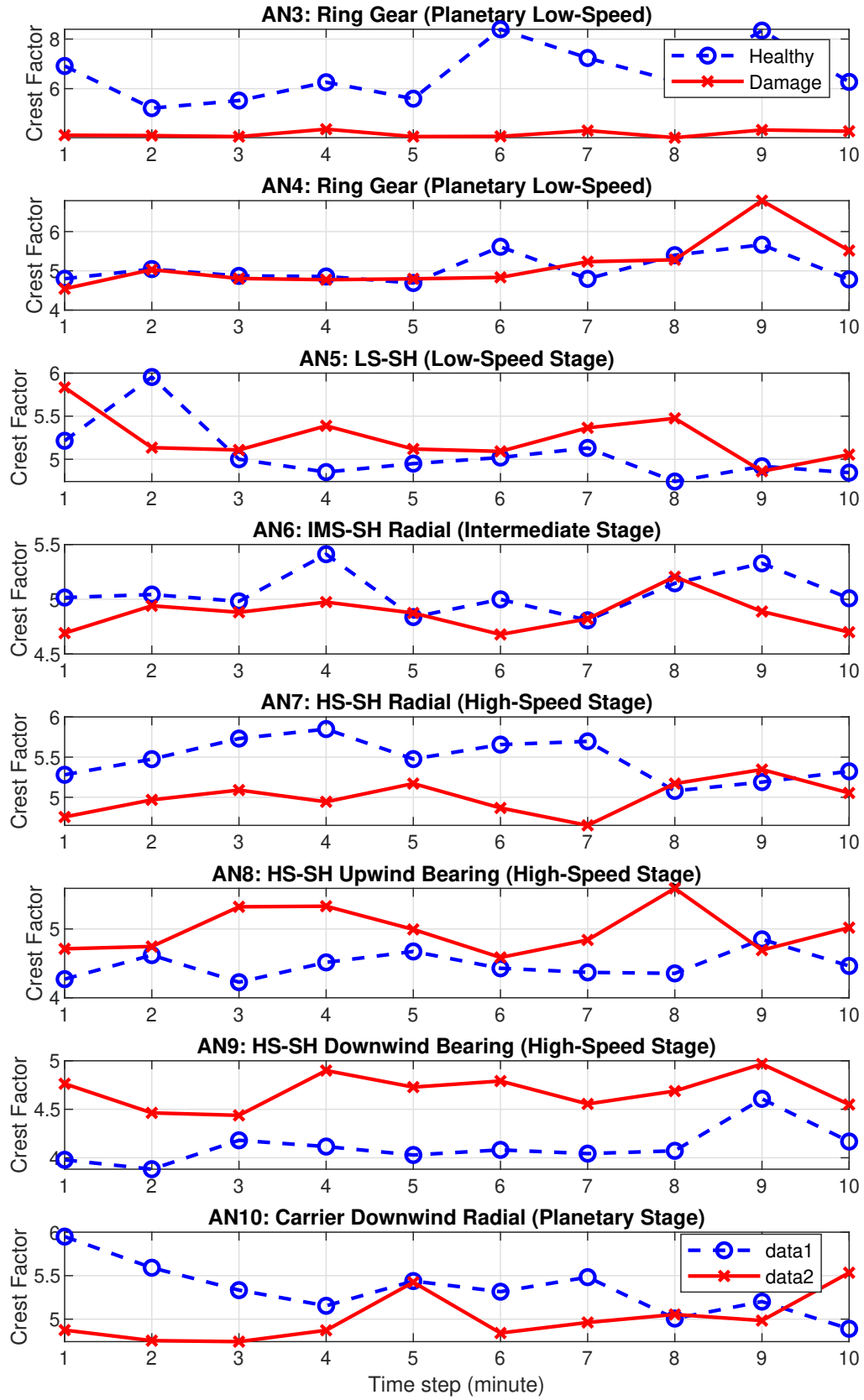


Figure 2: CF for all acceleration data (Healthy vs. Damaged)

(such as a single fractured tooth) are not present. Instead, the vibration profile was replaced by continuous, broadband friction noise across the contact surfaces. This specific trend matches a general symptom for gear tooth scuffing, micro-pitting, and surface polishing wear mechanism that removes sharp geometric impacts and increases continuous noise.

### 3 High-Frequency Noise Floor Assessment

To isolate micro-frictional shearing from low-frequency rotational variations (like shaft unbalance or running misalignments), a high-frequency noise floor assessment has been undertaken. Sensor data studied for this assessment include - AN3, AN4, AN5, AN6, and AN7.

Welch's Power Spectral Density (PSD) estimation has been applied for this case, using a moving 8192-sample Hanning window and a 50% overlap configuration. The integrated mean power density across the 10–18 kHz band was converted to a logarithmic decibel (dB) scale to track continuous energy shifts over the 10-minute timeline.

From Figure 3 (top-left), it is observed that the low-speed ring gear sensors (AN3 and AN4) exhibit divergent trends. For AN3 (6 o'clock position), the damaged trend line dropped below the healthy baseline while for AN4 (12 o'clock position), the damaged curve went upwards, compared to the 'healthy' reference. Further it should be noted that the trend corresponding to healthy and damaged data is different. The healthy data rise and fall for both sensors, while the damaged data mostly rises with time.

Further, for AN5, AN6, and AN7, a permanent offset from the healthy baseline has been observed for each of the cases (Figure 3).

These discrepancies points to a loss of structural stiffness or local component decoupling (especially for low-speed stage). In high-speed (AN7 and somewhat AN6) industrial drivetrains, energy accumulation within the 10–18 kHz band is a direct byproduct of lubrication film breakdown and subsequent metal-to-metal tearing (scuffing).

### 4 Planetary Gearbox Defect Detection Using FFT

To evaluate the low-speed planetary component anomalies observed by sensors AN3 and AN4, Fast Fourier Transform (FFT) has been used. 1 minute data from file 'D10.mat' has been used for this study here. The original 40 kHz signals were decimated (used in-built MATLAB function to resample data at a lower rate after lowpass filtering) by a factor of 40 to establish an optimized sampling rate of 1000 Hz. Hanning window function was applied to the downsampled signals to suppress spectral leakage, and a discrete FFT followed. The single-sided magnitude spectrum was converted to a decibel amplitude reference format, with the display noise floor locked at a threshold of -60 dB to focus strictly on the 0 to 150 Hz planetary operational zone.

In Figure 4, fine-dotted vertical vectors laid on the plots represent a  $\pm 3$  order cluster of fault sidebands surrounding the gear mesh centers for better readability. Analysis of both spectrum subplots revealed identical, high-amplitude resonance peaks lining up with the theoretical indicators.

The prominent sidebands corresponding to the three mesh harmonics confirm heavy amplitude and phase modulation patterns matching the planet passing frequency ( $i \times f_{\text{pass}}$ ). When planet teeth pass through a localized structural defect on a stationary ring

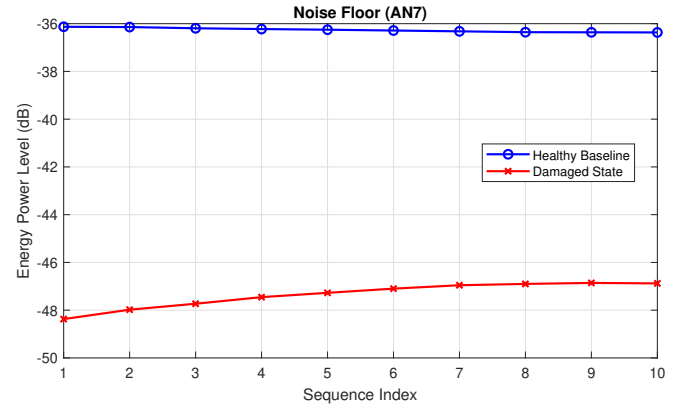
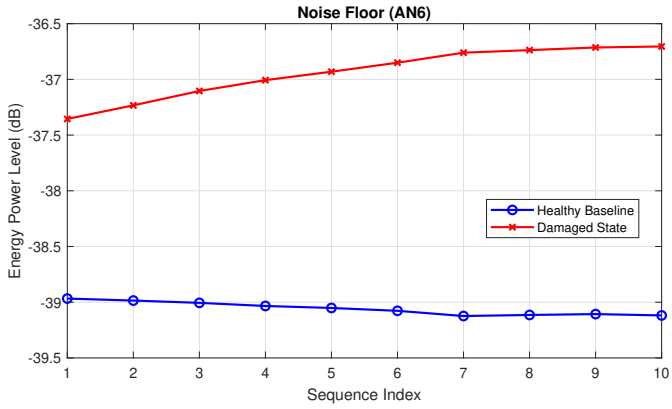
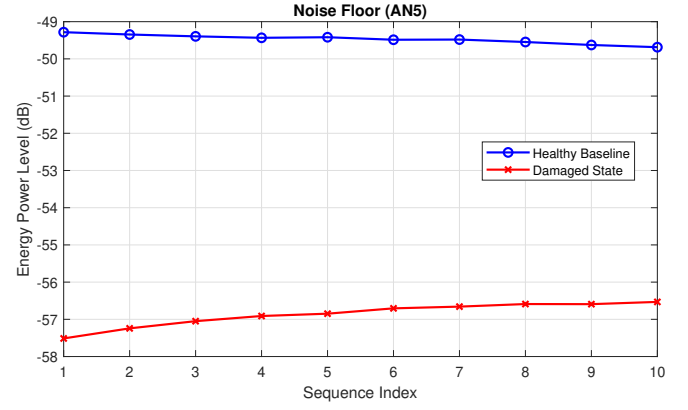
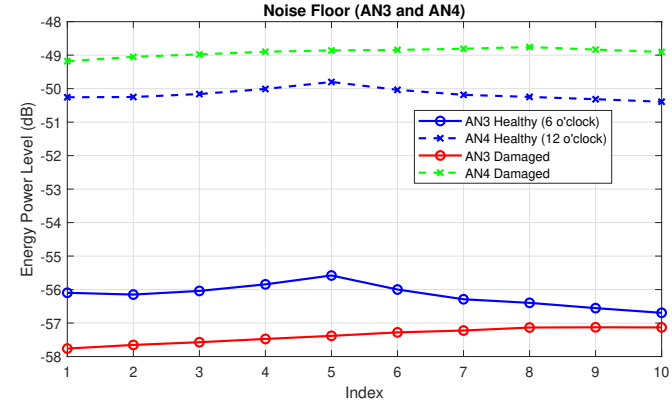


Figure 3: High-Frequency Noise Floor Trend Analysis

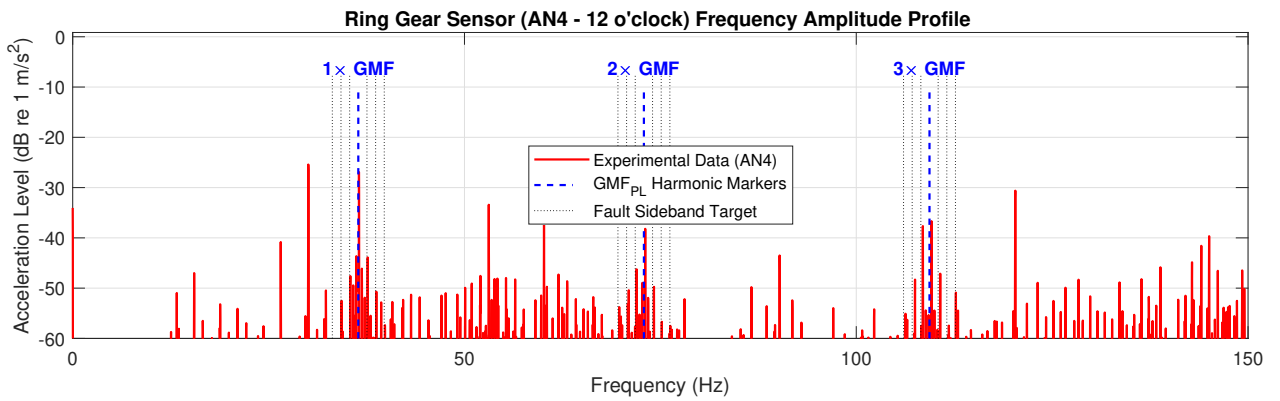
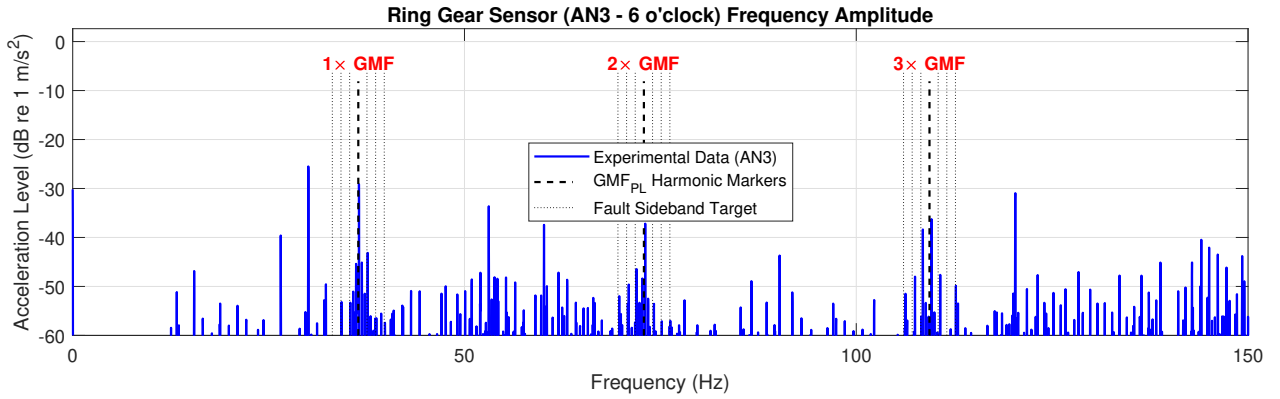


Figure 4: Planetary Gearbox Defect Detection - AN3 and AN4

gear, the meshing stiffness drops periodically every time a planet gear passes over the damaged zone. This mechanical modulation produces sidebands spaced exactly at the planet passing frequency. This is an indication of presence of structural ring gear defect - scuffing/fretting.

To quantify the structural defect severity, a diagnostic tracking function has been established (refer to the code) which isolated a narrow spectral integration band ( $\pm 0.2$  Hz) to compute the direct Sideband Amplitude Ratio (SAR) indicator.

## 5 Summary

Table 1 provides a summary of the structural defects identified by the condition monitoring method, mapping each fault to its primary sensor channel and signal processing technique.

Table 1: Summary

| Component                          | Primary Sensor | Algorithmic Processing Method                        | Type of Defect Identified                          | Assigned Severity & Diagnostic Reason                                                                                                                                             |
|------------------------------------|----------------|------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ring Gear</b>                   | AN3, AN4       | Decimated Low-Freq FFT & Logarithmic Scaling         | Localised Ring Gear Scuffing and Fretting          | <b>CRITICAL</b><br>Sideband Amplitude Ratio (SAR) exceeding the baseline limit.                                                                                                   |
| <b>Planetary Assembly Casing</b>   | AN3, AN4, AN5  | Welch PSD Noise Floor Integration (10–18 kHz)        | Assembly/ Structural damage                        | <b>HIGH</b><br>Marked by asymmetric high-frequency shifts.                                                                                                                        |
| <b>Intermediate Parallel Stage</b> | AN6            | Time-Domain Trending & Welch Noise Floor Integration | Distributed Gear Tooth Polishing Wear and Fretting | <b>MEDIUM</b><br>Indicated by high-frequency rise and a dominant mesh harmonics.                                                                                                  |
| <b>High-Speed Parallel Stage</b>   | AN7            | Time-Domain Trending & Welch Noise Floor Integration | High-Speed Gear Set Pinion / Gear Scuffing         | <b>CRITICAL</b><br>Driven by a massive surge in high-frequency noise floor energy and strong sidebands. This confirms severe lubrication breakdown and destructive metal tearing. |

## 6 Conclusion

Global RMS tracking flagged active energy anomalies on sensors AN6 and AN7, while Crest Factor trends isolated a signature change on the planetary stage. High-frequency

noise floor tracking confirmed severe surface friction, and FFT analysis isolated the localized modulations. This workflow effectively identified localized high-speed gear scuffing from uniform intermediate stage polishing wear and planetary ring gear degradation.

The spectrum also identified prominent, unmodulated peaks outside the planetary mesh zones, particularly a high-amplitude spike near 53 Hz. This frequency does not match any planetary stage harmonics, pointing to secondary damage elsewhere in the drivetrain (e.g., intermediate shaft upwind/downwind bearing assembly damage or dents).

## References

- [1] Shawn Sheng. Wind turbine gearbox condition monitoring vibration analysis benchmarking datasets. Open Energy Data Initiative (OEDI), National Renewable Energy Laboratory, <https://doi.org/10.25984/1844194>, 2014. Accessed: 2026-06-16.